

CSC111 Fall 2025 Midterm Sample Exam Questions

1. For the following multiple-choice questions, each question has exactly one correct answer key. If multiple choices are correct, choose the option “All of the above”. **Fill in the answer keys in the table above.**
(Answer keys written in the question area will not be counted.)

MCPs will be based on the quizzes.

1. For the 4-bit binary values in the table below, show the equivalent decimal values when the data is interpreted as unsigned binary or signed binary.

Binary Value	Signed Decimal Value	Unsigned Decimal Value
0000	0	0
0001	1	1
0111	7	7
1000	-8	8

2. (a) Assume a 4-bit system. Copy bits 3..2 in R4 to be the right most bits of R5, and set other bits in R5 to 0. Use only LSL, LSR, AND instructions.
(b) Assume a 4-bit system. Copy bits 3..2 in R4 to be the left most bits of R5, and keep other bits in R5 unchanged. Use only LSL, LSR, AND, OR instructions.

ANS:

(a)

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AND R5, R4, #0b1100
LSR R5, R5, #2
```

(b)

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AND R3, R4, #0b1100
AND R5, R5, #0b0011
ORR R5, R5, R3
```

3. Fill in the table for the N, V flags and condition for signed comparison.

ANS:

Signed Compare Example	N	V	Condition (GE or LT)
%R1 = 0101, R2 = 0011 CMP R1, R2			
%R1 = 0111, R2 = 1101 CMP R1, R2			
%R1 = 0010, R2 = 0101 CMP R1, R2			
%R1 = 1001, R2 = 0011 CMP R1, R2			

ANS:

Signed Compare Example	N	V	Condition (GE or LT)
%R1 = 0101, R2 = 0011 CMP R1, R2	0	0	GE
%R1 = 0111, R2 = -3 1101 CMP R1, R2	1	1	GE
%R1 = 0010, R2 = 0101 CMP R1, R2	1	0	LT
%R1 = 1001, R2 = 0011 CMP R1, R2	0	1	LT

4. Assume a 5-bit system. Fill in the table for the N, V flags and condition for signed comparison.

ANS:

Signed Compare Example	N	V	Condition (GE or LT)
%R1 = 00111, R2 = 00011 CMP R1, R2			
%R1 = 00011, R2 = 00111 CMP R1, R2			
%R1 = 10110, R2 = 00111 CMP R1, R2			
%R1 = 01010, R2 = 11001 CMP R1, R2			

ANS:

Signed Compare Example	N	V	Condition (GE or LT)
%R1 = 00111, R2 = 00011 CMP R1, R2	0	0	GE
%R1 = 00011, R2 = 00111 CMP R1, R2	1	0	LT
%R1 = 10110, R2 = 00111 CMP R1, R2	0	1	LT
%R1 = 01010, R2 = 11001 CMP R1, R2	1	1	E

5. (20 points) Program understanding. Translate the following assembly program code into C and explain what it does.

Assembly program	C program
<pre> MOV R2, #1 MOV R1, #1 loop CMP R1, R0 BGT done MUL R2, R1, R2 ADD R1, R1, #1 B loop done MOV R0, R2 </pre>	

ANS: Factorial = R0!

Assembly program	C program
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	MOV R2, #1	int factorial(int n) {
	MOV R1, #1	int result = 1; // R2
loop	CMP R1, R0	int i = 1; // R1
	BGT done	while (i <= n) { // loop until i > ns
	MUL R2, R1, R2	result = result * i;
	ADD R1, R1, #1	i = i + 1;
	B loop	}
		return result; // R0 holds return value
done	MOV R0, R2	}